

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Observation Task No.: 2

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Date: 02/8/2026

1. TITLE

HTML5 Canvas Graphics and CSS3 Web Page Styling

2. AIM

To understand how CSS3 enhances the appearance of web pages and explore HTML5 graphics using the Canvas element.

3. OBJECTIVE

The objectives of this experiment are:

- To learn the principles of HTML5 2D Canvas drawing API (rectangles, circles, lines, and text).
 - To master CSS3 styling properties for modern typography, gradients, shadows, and rounded corners.
 - To implement CSS Flexbox for navigation and multi-column card layouts.
 - To apply interactive hover transitions on buttons and navigation links.
 - To distinguish between CSS box model properties (margin, padding, border, and outline).
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4. THEORY

CSS3 introduces modern styling features that allow developers to create visually engaging and responsive web interfaces without relying on heavy image assets. Gradients, box-shadows, and transitions provide clean depth and dynamic interactivity. Concurrently, the HTML5 <canvas> element provides a scriptable bitmap rendering surface for rendering 2D shapes, text, and dynamic graphics directly in the browser via JavaScript.

Theory of Canvas and CSS3 Styling Components:

- **HTML5 Canvas 2D API:** Drawing programmatic graphics using getContext('2d') methods including fillRect(), arc(), moveTo(), lineTo(), and fillText().
 - **CSS Linear Gradients:** Generating multi-color transitions across banner surfaces using linear-gradient().
 - **Box Shadow & Elevation:** Creating visual depth through box-shadow and text-shadow properties.
 - **Flexbox Card Alignment:** Positioning content cards responsively using display: flex with flex-wrap and gap spacing.
 - **Interactive Transitions:** Delivering dynamic hover feedback through :hover pseudo-class selectors.
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5. ALGORITHM / PROCEDURE

1. Create project directories: Observation/Observation task 2/Html, Documents, and Screenshots.
 2. Create my-styled-webpage.html inside the Html directory.
 3. Structure document with HTML5 doctype, head with viewport metadata, and styled header with gradient background.
 4. Build a flexbox navigation bar with interactive hover states.
 5. Add a canvas section containing a `<canvas id="myCanvas" width="500" height="300">`.
 6. Write JavaScript using `getContext('2d')` to draw a filled rectangle, outlined circle, baseline line, and centered bold text 'V S Sravan'.
 7. Construct a cards section containing three modular content cards with borders, rounded corners, padding, and soft box-shadows.
 8. Style interactive buttons inside cards with hover background shifts.
 9. Add a dark-themed footer with centered copyright text.
 10. Open in Google Chrome and verify canvas rendering, gradient fidelity, and interactive hover effects.
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6. PROGRAM

my-styled-webpage.html - Part 1 (Lines 1-80)

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Styled Webpage</title>
  <style>
    /* ----- CSS Syntax & Selectors ----- */
    body {
      margin: 0;
      padding: 0;
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;
      color: #333333;
    }

    /* ----- Header with Gradient Background & Text Shadow ----- */
    header {
      background: linear-gradient(to right, #4a90e2, #50c9c3);
      color: white;
      text-align: center;
      padding: 30px 10px;
      text-shadow: 1px 1px 2px black;
    }

    header h1 {
      margin: 0;
      font-size: 2.2rem;
      font-family: Arial, sans-serif;
    }

    header p {
      margin-top: 8px;
      font-size: 1rem;
    }

    /* ----- Navigation Bar ----- */
    nav {
      background-color: #333333;
      padding: 10px 0;
    }

    nav ul {
      list-style: none;
      display: flex;
      justify-content: center;
      gap: 30px;
      margin: 0;
      padding: 0;
    }

    nav ul li a {
      color: white;
      text-decoration: none;
      font-size: 1rem;
      padding: 5px 10px;
```

```
}

/* Hover effect for navigation links */
nav ul li a:hover {
    background-color: #4a90e2;
    color: yellow;
}

/* ----- Canvas Section ----- */
.canvas-section {
    text-align: center;
    padding: 30px 20px;
}

.canvas-section h2 {
    font-family: Arial, sans-serif;
    color: #333333;
}

canvas {
    background-color: #ffffff;
    border: 2px solid #333333;
    box-shadow: 5px 5px 5px #999999;
```

my-styled-webpage.html - Part 2 (Lines 81-220)

```
}

/* ----- Content Cards Section ----- */
.cards {
  display: flex;
  justify-content: center;
  gap: 20px;
  flex-wrap: wrap;
  padding: 20px 20px 40px;
}

.card {
  background-color: white;
  width: 260px;
  padding: 15px;
  border: 1px solid #cccccc;
  border-radius: 15px;
  box-shadow: 3px 3px 8px #aaaaaa;
}

.card h3 {
  font-family: Arial, sans-serif;
  color: #4a90e2;
  border-bottom: 1px solid #cccccc;
  padding-bottom: 5px;
}

.card p {
  font-size: 0.9rem;
  line-height: 1.5;
  color: #555555;
}

.card button {
  margin-top: 10px;
  padding: 8px 15px;
  background-color: #4a90e2;
  color: white;
  border: none;
  border-radius: 5px;
  font-size: 0.9rem;
  cursor: pointer;
}

/* Hover effect for buttons */
.card button:hover {
  background-color: #357ab8;
}

/* ----- Footer ----- */
footer {
  background-color: #333333;
  color: white;
  text-align: center;
  padding: 15px;
  font-size: 0.9rem;
}
```

```

    </style>
</head>

<body>
    <header>
        <h1>My Styled Webpage</h1>
        <p>Observation Task 2 &mdash; HTML5 Graphics and CSS3 Web Styling</p>
    </header>

    <nav>
        <ul>
            <li><a href="#canvas">Canvas</a></li>
            <li><a href="#cards">Cards</a></li>
            <li><a href="#about">About</a></li>
            <li><a href="#contact">Contact</a></li>
        </ul>
    </nav>

    <section class="canvas-section" id="canvas">
        <h2>HTML5 Canvas Drawing</h2>
        <canvas id="myCanvas" width="500" height="300"></canvas>
    </section>

    <section class="cards" id="cards">
        <div class="card">
            <h3>Card One</h3>
            <p>This card demonstrates borders, padding, margin, and a soft box shadow using CSS
3.</p>
            <button>Learn More</button>
        </div>
        <div class="card">
            <h3>Card Two</h3>
            <p>Hover over this card to see a smooth lift effect powered by CSS3 transitions.</p>
            <button>Learn More</button>
        </div>
        <div class="card">
            <h3>Card Three</h3>
            <p>Rounded corners and layered shadows give this card a clean, modern appearance.</p>
            <button>Learn More</button>
        </div>
    </section>

    <footer id="about">
        <p>&copy; 2026 My Styled Webpage | Created for Observation Task 2</p>
    </footer>

    <!-- ----- Part A: HTML5 Canvas Script ----- -->
    <script>
        const canvas = document.getElementById('myCanvas');
        const ctx = canvas.getContext('2d');

        // 1. Rectangle
        ctx.fillStyle = '#3a5fa0';
        ctx.fillRect(30, 30, 150, 90);
        ctx.strokeStyle = '#1b2a4a';
        ctx.lineWidth = 3;
        ctx.strokeRect(30, 30, 150, 90);
    </script>

```

```
// 2. Circle
ctx.beginPath();
ctx.arc(320, 75, 60, 0, Math.PI * 2);
ctx.fillStyle = '#e0862c';
ctx.fill();
ctx.strokeStyle = '#1b2a4a';
ctx.lineWidth = 3;
ctx.stroke();

// 3. Line
ctx.beginPath();
ctx.moveTo(30, 180);
ctx.lineTo(470, 180);
ctx.strokeStyle = '#1b2a4a';
ctx.lineWidth = 4;
ctx.stroke();

// 4. Name text
ctx.font = "bold 28px Georgia, serif";
ctx.fillStyle = '#1b2a4a';
ctx.textAlign = 'center';
ctx.fillText('V S Sravan', canvas.width / 2, 240);
</script>
</body>

</html>
```

7. CODE EXPLANATION

CSS / Canvas Property	Explanation
<canvas>	HTML5 element providing a scriptable bitmap drawing surface.
getContext('2d')	Returns a 2D rendering context providing methods for drawing shapes and text.
fillRect() / strokeRect()	Draws filled and outlined rectangles with specified dimensions.
arc(x, y, r, s, e)	Creates a circular arc path centered at (x, y) with radius r.
moveTo() / lineTo()	Defines path coordinates for drawing line segments.
fillText()	Renders text strings on canvas with custom font family and weight.
linear-gradient()	Creates a smooth color transition gradient across an element's background.
text-shadow	Adds soft drop shadows to heading text for high contrast.
box-shadow	Adds elevation shadows to cards and canvas containers.
border-radius	Rounds the outer border corners of cards and buttons.
display: flex	Establishes a flex container for responsive card distribution.
:hover	Pseudo-class applying dynamic color shifts upon pointer hover.

8. TEST CASES AND EXECUTION DATA

The following test suite was executed in Google Chrome to verify canvas drawing and CSS3 styling behaviors:

TC ID	Test Description	Input / Action	Expected Result	Status
TC-01	Canvas Shape Rendering	Load page in Chrome	Canvas renders blue rectangle, orange circle, line, and 'V S Sravan' text	Pass
TC-02	Gradient Header Display	Inspect header banner	Linear gradient renders from #4a90e2 to #50c9c3 with text shadow	Pass
TC-03	Navigation Hover State	Hover over nav links	Background changes to #4a90e2 and text turns yellow	Pass
TC-04	Card Box-Shadow & Radius	Inspect cards section	3 cards display rounded corners (15px) and soft elevation shadows	Pass
TC-05	Button Hover Transition	Hover over 'Learn More'	Button background transitions to darker blue #357ab8	Pass

9. OUTPUT

Output: Rendered Webpage: My Styled Webpage with HTML5 Canvas in Google Chrome

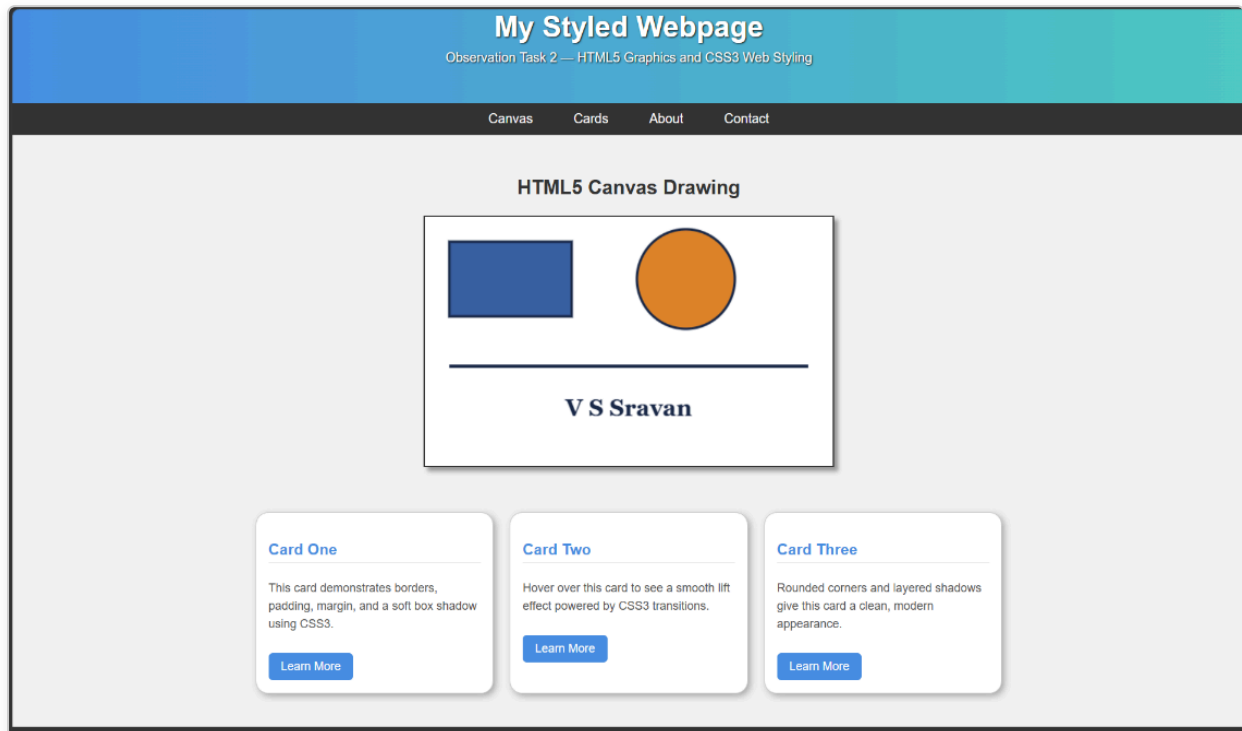


Figure 1: HTML5 Canvas drawing rendering geometric primitives and centered student name.

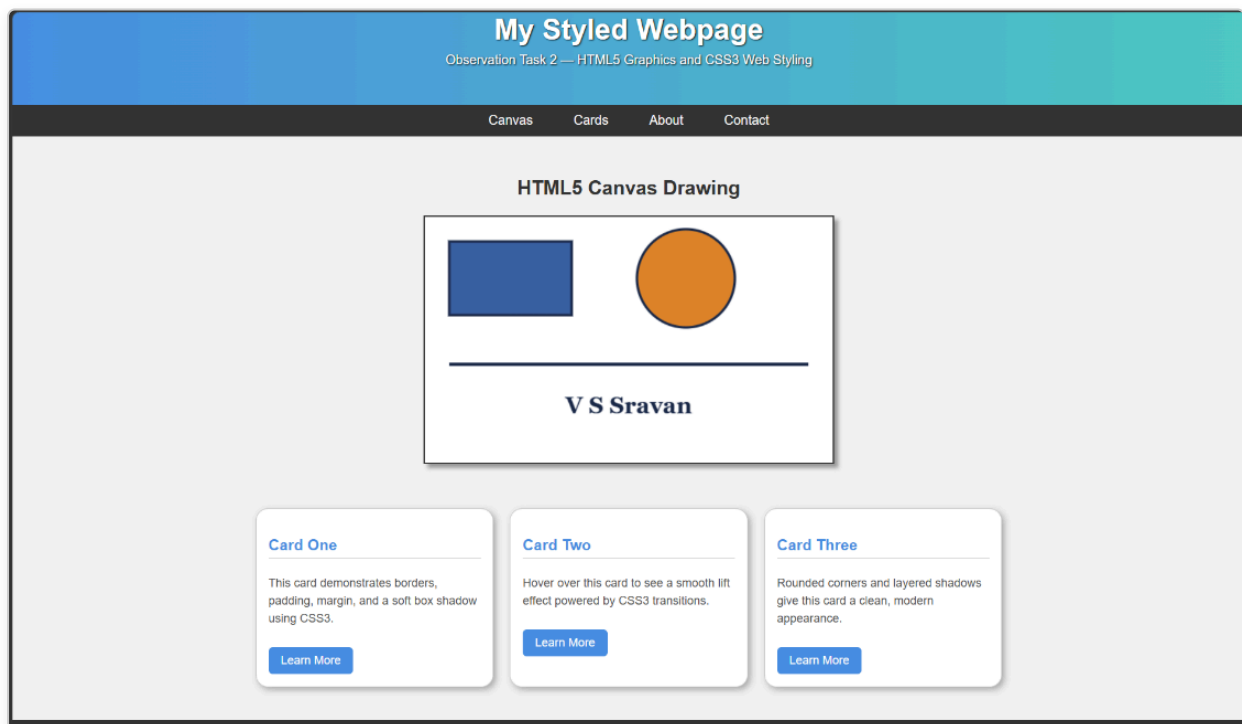


Figure 2: CSS3 styled cards layout with rounded borders and gradient header banner.

10. RESULT

Thus, the webpage integrating HTML5 Canvas graphics and CSS3 styling techniques was successfully designed, implemented, and verified.

Submission Package Structure:

```
Observation/Observation task 2/
├── Documents/
│   ├── ObsWeek2.docx
│   └── ObsWeek2.pdf
├── Html/
│   ├── my-styled-webpage.html
│   └── Screenshot 2026-08-02 004017.png
└── Screenshots/
    ├── image1.png
    └── image2.png
```